CVMU

Vallabh Vidyanagar

PROGRAM_LIST_ CA&IT_PRACTICALS--Sem 3

(Syllabus Effective from June 2021)

Assignment – 1

[Class and Object]

1.	**Write a program to generate the following series (using class fibo) :** **1 1 2 3 5 8 …..**
2.	Write a program to print factorial of N. (Using class fact)
3.	**Build a class person, which contains person name, age and city as data members and member functions getdata( ) and printdata ( ). Write a C++ program to read and display a person data.**
4.	**Define a class student, which accept student number, name and marks of three subjects. Calculate total and percentage using different member functions. (For single student only)** **Output:** **Student Number : ……** **Student Name : ……** **Marks 1 : …..** **Marks 2 : …..** **Marks 3 : …..** **Total : ….** **Percentage : …..**
5.	**Write a C++ program to build a class myarray having integer array as a data member and following member function.** **[i]. void sort_asc() : To sort an array in ascending order.** **[ii] void sort_desc() : to sort an array in descending order.** **Write menu driven program for sorting either ascending or descending order and also for exit from a program.**

6.	Write a C++ program to build a class String and write down the following member functions for string manipulation. [i]. strlen1() : To count the length of the string. [ii]. strcpy1() : To copy one string to another. [iii]. strcat1() : To combine two string into one. [iv]. strcmp1() : To compare two string.
7.	**To make a class string, which contain a character array. Write a C++ program to display the menu as shown below and perform the task according to the menu.** **MENU** **-----------------------------** **1. Reverse the String** **2. Convert the String to uppercase** **3. Convert the String to lowercase** **4. Print the Length of the String** **5. Exit** **Enter your choice…**

<u>Assignment – 2</u>

[Constructor]

1.	**Create a class EMP having data members like employee number, name and basic salary. Initialize its members by constructor and display it.**
2.	**Write a C++ program that reads an integer number, display the menu as follows and perform according to choice. (Use the constructor to initialize the value of variables)** **MENU** **------------------------------** **1. Sum of first N odd Numbers** **2. Sum of first N even Numbers** **3. Exit**

3.	**Write a C++ program to interchange two-integer number and two-real number without using the third variable. (Use concept of constructor overloading)**
4.	**Write a C++ program that reads an integer number, display the menu as follows and perform according to choice. (Use constructor to initialize the values)** **MENU** **1. Factorial using For…Next** **2. Factorial using Do…While** **3. Factorial using While** **4. Exit**

Assignment – 3

[Array of Object]

1	Create a class BANK to represent a bank account. Include the data members like name of depositor, account no., type of a/c (Saving, Current, FD), and balance amount. Make the function (1) To assign initial values (2) To deposit an amount (3) To withdraw an amount after checking balance (4) To display name and balance.
2	**Create a class cricketer having data members as: name, age, for which country he plays, type of cricketer (batsman/bowler), total matches he has played, total runs, total wickets he has taken.** **Member functions :** **(1) Assign initial values (2) To enter data (3) To display the data.** **Write a C++ program for 5 cricketers.**
3	**Write a C++ program that creates a class named salesman and read the following information for N number of salesman of a company. 1. Salesman no. 2. Quantity sold (above 50) 3. Rate per Quantity. Calculate amount &**

commission for each salesman according to following rules. Also calculate total commission.

Amount = Quantity Sold * Rate

if amount <= 1000 commission rate = nil .

for next 1000 amount comm. rate = 15%

for next 2000 amount comm. rate = 20%.

for above 4000 amount comm. rate = 25%.

Print information as follows.

ABC COMPANY

Salesman Information

Salesman no.	Quantity sold	Rate	Amount	Commission
TOTAL	------		-----	------

4 Write a C++ program for the following that create a class called employee and read the employee number, name, basic salary. Calculate and print individual pay slip for N employee as shown below. Also print employee name, gross salary of highest gross salary. Use following rules for calculation.

DA calculation : for basic <= 1500 da = 159% of basic.

for 1500 < basic <= 2500 then da = 148% of basic.

for 2500<basic<=4500 then da = 125% of basic.

for 4500<basic then da= 90% of basic

HRA calculation: code = 0 hra not applicable(hra=0)

code=1 then hra=120

code=2 then hra=250

code=3 then hra=400

	Gross Salary = da+hra+basic Income tax = 10% of gross salary. Professional tax = 80 Rs. Total Deduction = income tax + professional tax Net Salary = Gross Salary - Total Deduction. EMPLOYEE PAYSLIP Employ Number : ------- Name : -------- --- Earnings Amount Deductions Amount BASIC ------- DA ------- PT -------- HRA ------- INCOME TAX -------- GROSS ------- TOTAL DEDUCTION -------- --- NET SALARY --------
5	Write a C++ program that read the Name, Roll no, Marks of BCA201, BCA202…to… BCA210, Sex Code (1 for Male and 2 for Female) for students of S.Y.BCA and Print the mark sheet for each student. Also print the total no. of pass male and female students. Maximum mark of each subject is 100. Passing standard is 35%. If student is fail in more then two subject then declare result is "FAIL" If student is fail in one or two subject then declare result is "ATKT" If student is pass in all subject then declare the student as PASS. Award class only for those student whose result is pass

If percentage < 48 then class is "PASS"

If 48 <= percentage < 60 then class is "SECOND"

If 60 <= percentage < 70 then class is "FIRST"

If percentage >= 70 then class is "DISTINCTION"

Student Marksheet

##############

```
Number   Name    Sub1   Sub2   Sub3    Sub4    Sub5   Sub6  Total  Per  Result
---------    --------   -------  ------   ------    -----    ------   -----   -------  ---  -------
xxx       .....
.
.
---------    --------   -------  ------   ------    -----    ------   -----   -------  ---  -------
```

Sr. No.	Result	Male	Female	Total
1	Pass	xx	xx	xx
2	Fail	xx	xx	xx
3	ATKT	xx	xx	xx

6 Declare a class called STUDENT, which contains data members as: roll number, name, and marks of six subjects. Write a program to read the data members and calculate total, percentage and result of them. Display mark sheet of all students as shown below. Passing criteria is 40 Marks. Maximum marks of each subject are 100.

If student pass in all subject then result is "PASS".

If student fail in one or two subject then result is "ATKT".

If student failed in more than two subject then result is "FAIL" .

Student Marksheet

##############

```
Number   Name    Sub1   Sub2   Sub3    Sub4    Sub5   Sub6  Total  Per  Result
---------    --------   -------  ------   ------    -----    ------   -----   -------  ---  -------
```

	xxx . . . . ---------- -------- ------- ------ ------ ----- ------ ----- ------- --- ------- Total student : ____ Total Pass : ____ Total Fail : ____ Total ATKT : _____ (use array of object)

Assignment – 4

[Inheritance]

1	**Create a class STUDENT having data members as student number, student name and array of five subject marks. Maximum marks of each subject is 100 marks. Derived a class RESULT from the above class STUDENT having data members as total, percentage and result(grade). Note that the passing criteria is 30 marks of each subject. Define a members function calculate() that calculates total, percentage and grade.** **The criteria for grade calculation:** **If percentage >=70% then grade 'A'** **If percentage < 70% and >=60% then grade 'B'** **If percentage < 60% and >=50% then grade 'C'** **If percentage < 50% and >=30% then grade 'D'** **If percentage < 30% then grade 'FAIL'** **Define a member function display() that prints the result of the N number of student in the following format:** **MARKSHEET**

	NUMBER NAME M1 M2 M3 M4 M5 TOTAL % RESULT **======= ===== == == == == == ===== == ======** **1** . . . . **2** . . . . **3** . . . . . **:** **:** . . . .
2	Create a class publisher to store the attributes publisher name, address and phone number. Derive a subclass book of publisher class to store the attributes book name, ISBN, author name and price. Also create 1. Parameterized constructor in the parent class to initialize the data members of the parent class. 2. Parameterized constructor in the child class to initialize the data members of the child class as well as the data members of the parent class, call the parent constructor with the child constructor. 3. A friend function in child class to display the formatted information of both parent and child attributes. Take input in the main method and create at least three objects of the book class in the main method. Display the book information.
3	**Create a class STUDENT having data members as student number, student name. From the above class derive another class MARKS that contains data member as an array of five subject marks. Maximum marks of each subject is 100 marks. Derived a class RESULT from the above class MARKS having data members as total, percentage and result(grade). Note that the passing criteria is 30 masks of each subject. Define a members function calculate() that calculates total, percentage and grade.** **The criteria for grade calculation:** **If percentage >=70% then grade 'A'** **If percentage < 70% and >=60% then grade 'B'** **If percentage < 60% and >=50% then grade 'C'**

	If percentage < 50% and >=30% then grade ‘D’ If percentage < 30% then grade ‘FAIL’ Define a member function display() that prints the result of the N number of student in the following format: MARKSHEET NUMBER NAME M1 M2 M3 M4 M5 TOTAL % RESULT ======= ===== == == == == == ===== == ====== 1 . . . . 2 . . . . 3 . . . . . : : . . . .
4	Define a class ACCOUNT having account information, account number and name of the account holder along with appropriate constructors to read these information. Derive a class TYPE from the above class with appropriate constructors to get the account type (Savings or Current). Read and display the account information for at least three accounts.
5	Imagine a publishing company that markets both book and audio-cassette versions of its works. Create a class publication that stores the title (a string) and price of a publication. From this class derived two classes: book and tape: a class book which adds a page count (type int) and tape which adds a playing time in minutes (type float). Each of this classes should have a getdata() function to to get its data from the user at the keyboard and a putdata() function to display its data. Write a main() program to test the book and tape classes by creating instances of them, asking the user to fill in their data with getdata() and then displaying the data with putdata().

Assignment – 5

[Function/Operator Overloading]

Function Overloading

1	**Write a C++ program for overloading a function "volume" to calculate the volume of cube, cylinder and a rectangular box** **where volume of** **Cube=a*a*a** **Cylinder= 3.14*r*h** **Rectangular box=l*b*h**
2	Create two member functions max() to find out maximum number among two numbers and three numbers. (overloading concept)
3	**Create overloaded function reverse() to find reverse of a string and integer number.**

Operator Overloading

1	**Construct a class Month with data members as month number. Write down overloaded operator++ for the increment the month number by one. Write a member function read() and display() to read a number of month and print a number of month.** **Input:** **Enter month number : 8** **Output:** **Next Month number is : 9** **Input:** **Enter month number : 12** **Output:** **Next Month number is : 1**
[2]	**Write a Class vector including data member as integer array of length five and member functions as:** **[a]. Parameterized constructor initialize data member** **[b]. Overloaded the + operator to add two vector and returning the result as vector.**

	[c]. Overloaded the - operator to subtract two vector and returning the result as vector. **[d]. Overloaded the * operator to perform scalar multiplication and returning the result as vector.** **[e]. A method to display the value of the vector.**
[3]	Build a class string. Use overloaded operator + to combine two string.
[4]	Build a class string. Use overloaded operator == to compare two string.
[5]	Build a class string. Use overloaded operator = to copy one string into another.
[6]	Construct a class Month with data members as month name. Write down overloaded operator++ for the increment the month name by one. Write a member function read() and display() to read a name of month and print a name of month. Input: Enter name of a month : March Output: Next Month name is : April
[7]	**Construct a class CALENDAR with data members as date, month and year. Implement constructors, overloaded constructors and destructors for the same. Write down overloaded operator ++ that will increment the date by one. Write a function display() to print data, month and year. Consider all situations.** **e.g. if date is 31/7/2005 then output : tomorrow is 1/8/2005**
[8]	Construct a class Month with data members as month number. Write down overloaded operator - - for the decrement the month number by one. Write a member function read() and display() to read a number of month and print a number of month. Input: Enter month number : 8 Output: Previous Month number is : 7 Input:

	Enter month number : 1 Output: Previous Month number is : 12
[9]	Construct a class Month with data members as month name. Write down overloaded operator - - for the previous month name. Write a member function read() and display() to read a name of a month and print a name of month. Input: Enter name of a month : June Output: Next Month name is : May
[10]	Construct a class Time with data members as hours, minute and second. Implement constructors, overloaded constructors and destructors for the same. Implement a overloaded operator ++ which will increment the time by one second. Write a function display() to print hours, minute and second. Assume that 24 hours clock.

****Note : Program definitions with Bold effect are to be written in Journal.**